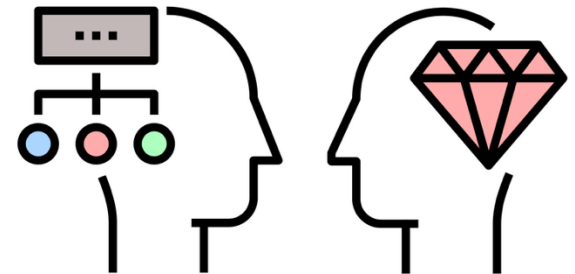


Machine Learning for Materials

9. Generative Artificial Intelligence

Aron Walsh

Department of Materials
Centre for Processable Electronics



Module Contents

1. Introduction
 2. Machine Learning Basics
 3. Materials Data
 4. Crystal Representations
 5. Classical Learning
 6. Deep Learning
 7. Building a Model from Scratch
 8. Accelerated Discovery
 - 9. Generative Artificial Intelligence**
 10. Future Directions
-

Key Concept #9

Generative AI learns and samples a data distribution

$$\begin{array}{ccc} & \textit{Probability} & \\ \text{sample } \mathbf{x} & \sim & p_{\theta}(\mathbf{x}|\mathbf{c}) \\ \textit{Stochastic sample} & & \textit{Condition} \\ \text{e.g. material} & & \text{e.g. target, constraints} \end{array}$$

Class Outline

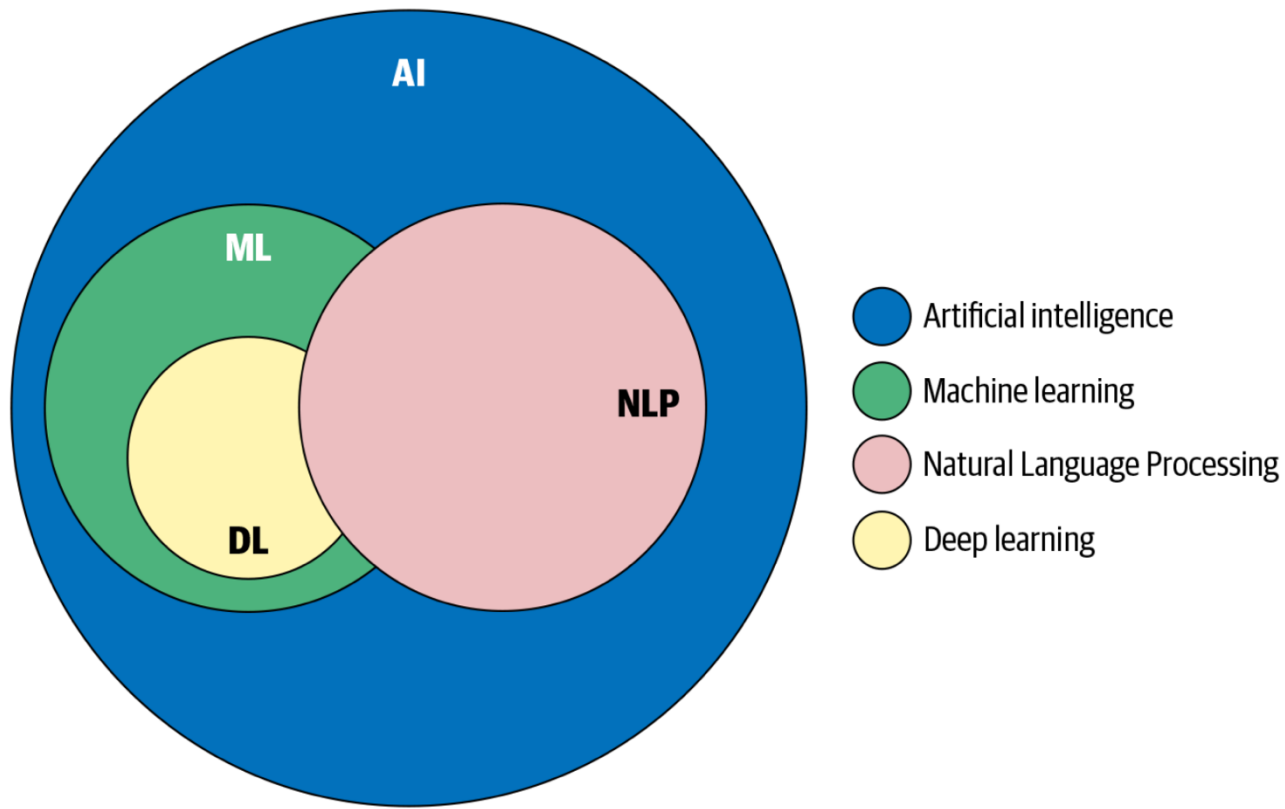
Generative AI

A. Large Language Models

B. From Latent Space to Diffusion

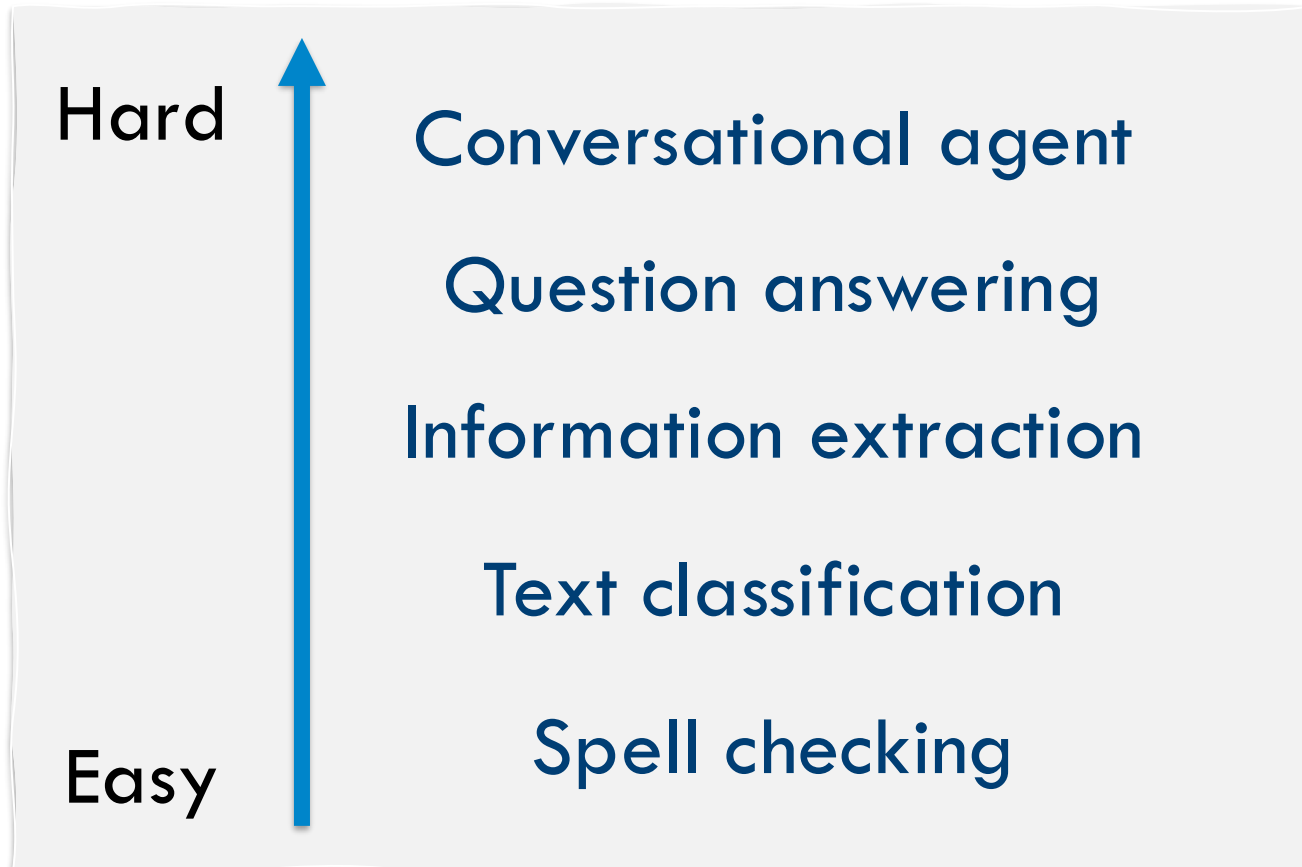
Natural Language Processing (NLP)

Branch of AI that focuses on the interaction between computers and human language



Natural Language Processing (NLP)

Branch of AI that focuses on the interaction between computers and human language



Language Models

Predictive text

I love materials because

of	shape	strong
they	are	essential
their	like	beautiful

Top words
ranked by
probability

“Temperature” of the text choices

I love materials because they ignite a symphony
of vibrant colors, tantalizing textures, and
wondrous possibilities that dance in the realms
of imagination, transcending boundaries and
embracing the sheer beauty of creation itself.

Sampling the
distribution
of probabilities
(“creativity”)

I love materials because they are essential.

Language Models

Large refers to the size and capacity of the model.
It must sample a (literary) combinatorial explosion

10^4 common words in English
 10^8 two-word combinations
 10^{12} three-word combinations
 10^{16} four-word combinations

**Language must be represented numerically
for machine learning models**

Tokens: discrete integer ID for each word (or subword unit)

Embeddings: dense continuous vector for each token

Text to Tokens

Example: “*ZnO is a wide bandgap semiconductor*”

Tokens

9

Characters

35

Note that Zn is
split into two
tokens
(not ideal for
chemistry)

ZnO is a wide bandgap semiconductor

Token-IDs

[57, 77, 46, 374, 3094,
4097, 43554, 39290, 87836]

768 dimensional embeddings are looked up from the (contextual) embedding matrix. These are model specific

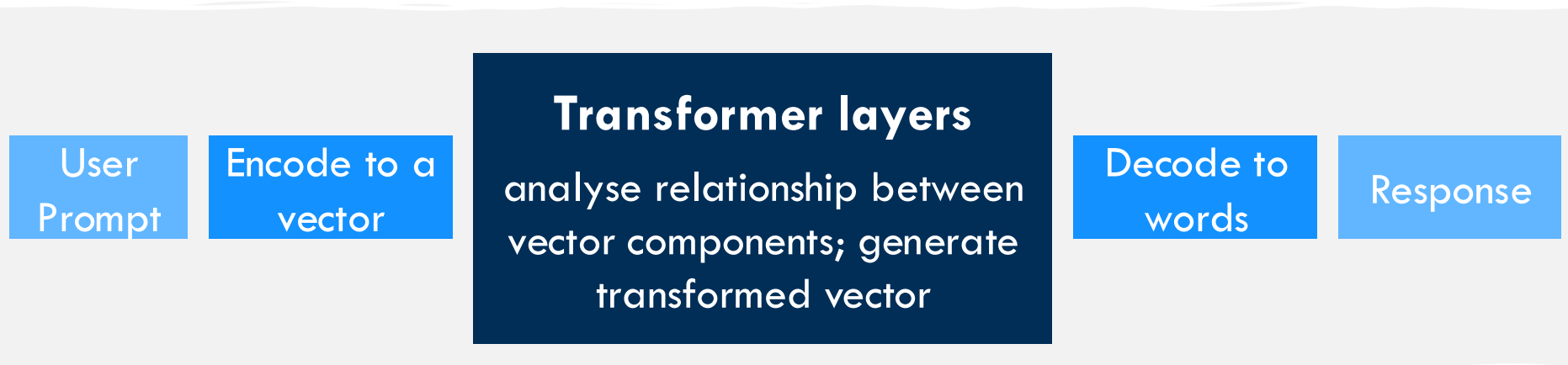
Large Language Models

GPT = “Generative Pre-trained Transformer”

Generate
new content

Trained on a
large dataset

Deep learning
architecture



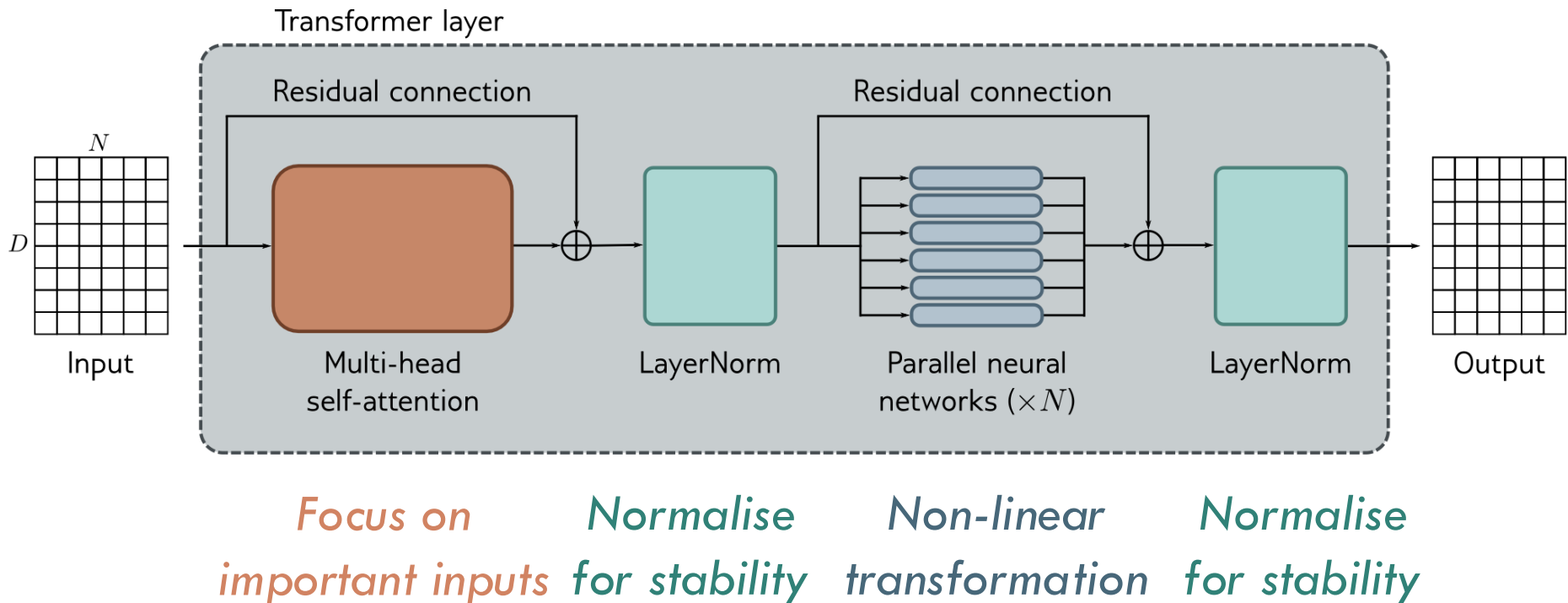
Key components of a transformer layer

Self-attention: smart focus on different parts of input

Feed-forward neural network: capture non-linear relationships

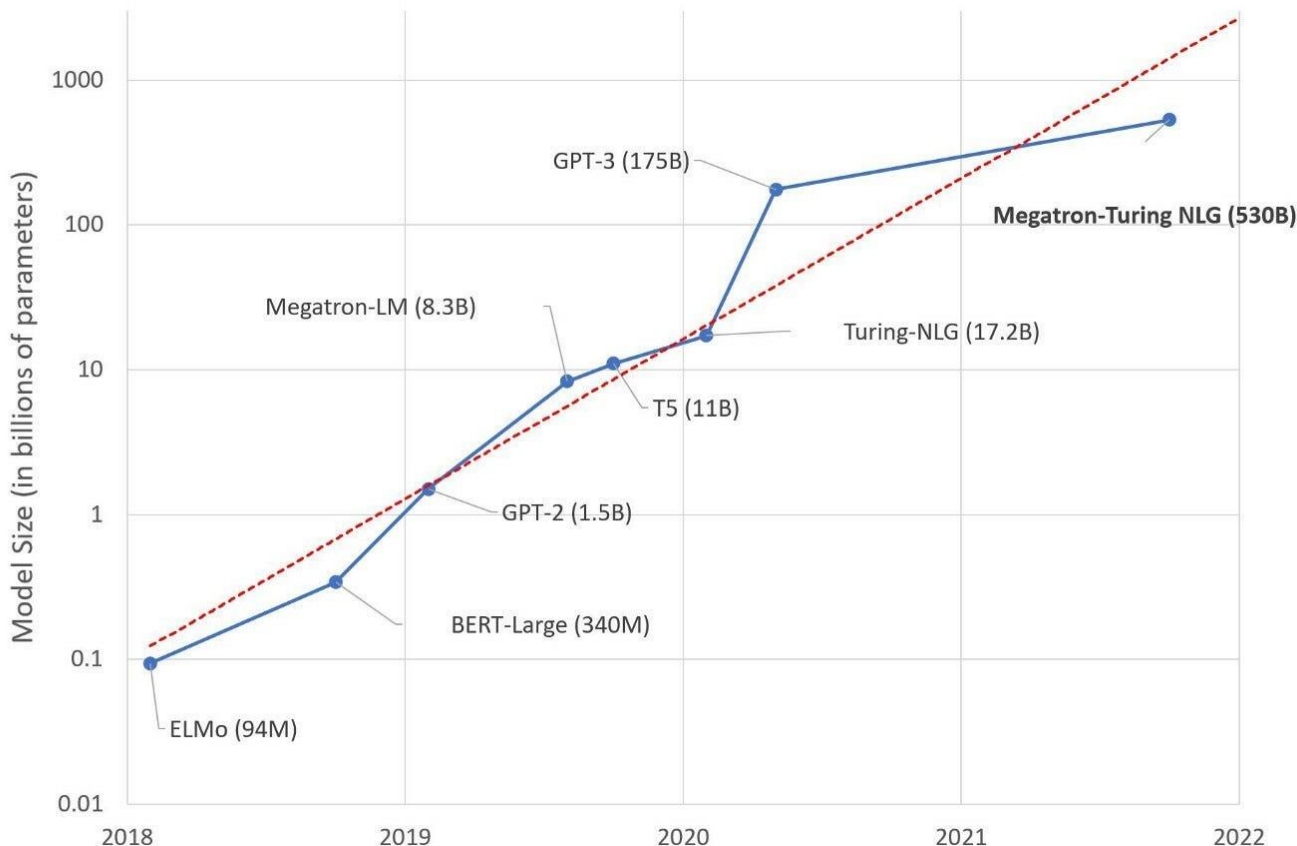
Large Language Models

Ongoing analysis into the physics of transformer architectures, e.g. rapid identification of correlations



Large Language Models

Deep learning models trained to generate text
e.g. BERT (370M, 2018), GPT-4 ($>10^{12}$, 2023)



Recent models
include:

Llama-4
(Meta, 2025)

Gemini-3
(Google, 2025)

GPT-5
(OpenAI, 2025)

PanGu-5.5
(Huawei, 2025)

Large Language Models

Essential ingredients of GPT and related models

Diverse
data

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

Deep
learning
model

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Validation
on tasks

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP ⁺ 20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

Large

Essential

Diverse
data

Deep
learning
model

Validation
on tasks

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



Share full article



1.3K

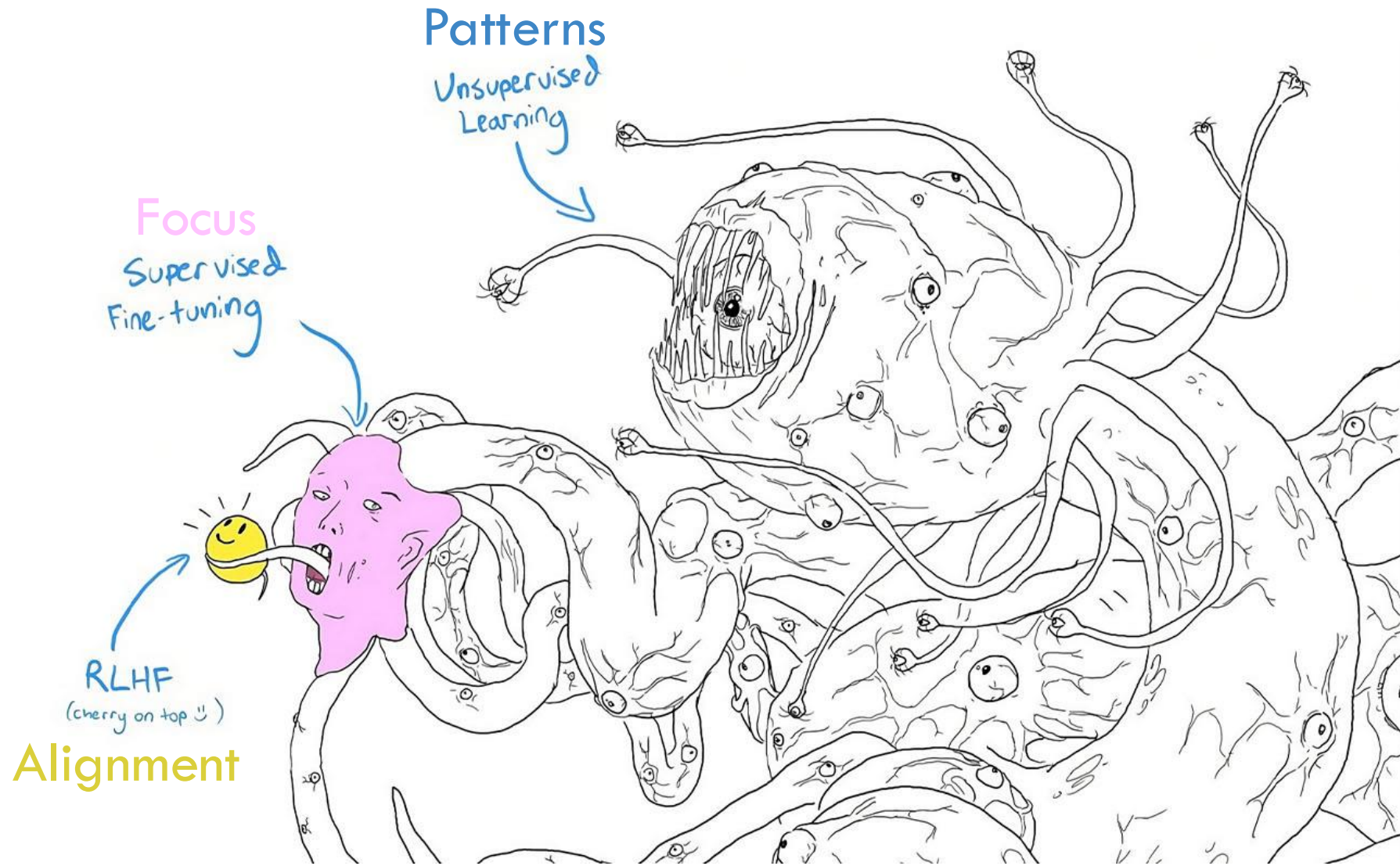


models

ing Rate

10^{-4}
 10^{-4}
 10^{-4}
 10^{-4}
 10^{-4}
 10^{-4}
 10^{-4}
 10^{-4}

Secret to Practical Success of LLMs



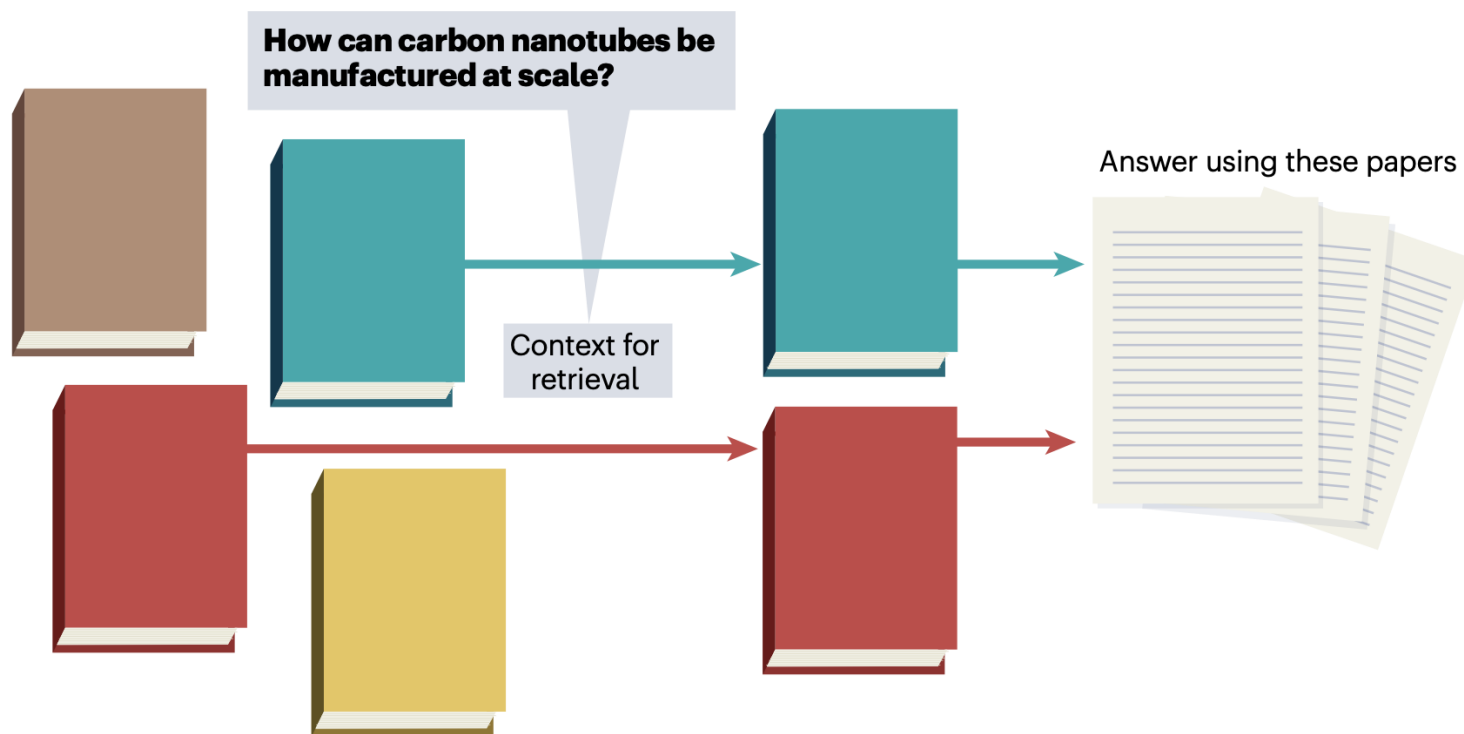
Large Language Models

What are the potential drawbacks and limitations of LLMs such as GPT?

- **Training data**, e.g. not up to date, strong bias
- **Context tracking**, e.g. limited short-term memory
- **Hallucination**, e.g. generate false information
- **Ownership**, e.g. fair use of training data
 - **Ethics**, e.g. appear human generated

LLMs for Materials

Many possibilities, e.g. read a textbook and ask technical questions about the content



LLMs for Materials

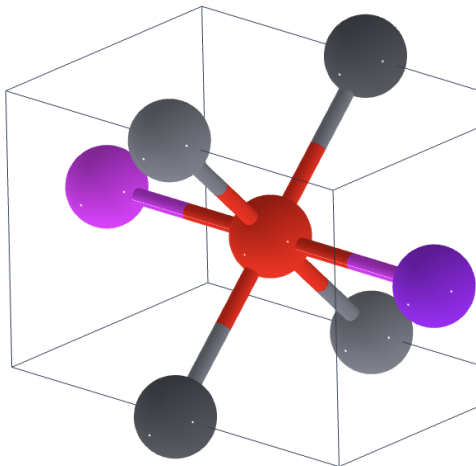
CrystaLLM: learn to write valid crystallographic information files (cifs) and generate new structures

Generate a crystal structure from a composition *

Composition:

optional optional

► Advanced options



- CIF (Symmetrized)
- CIF
- POSCAR
- JSON
- Prismatic
- VASP Input Set (MPRelaxSet)

LLMs for Materials

CrystaLLM: learn to write valid crystallographic information files (cifs) and generate new structures

Training set 2.2 million cifs

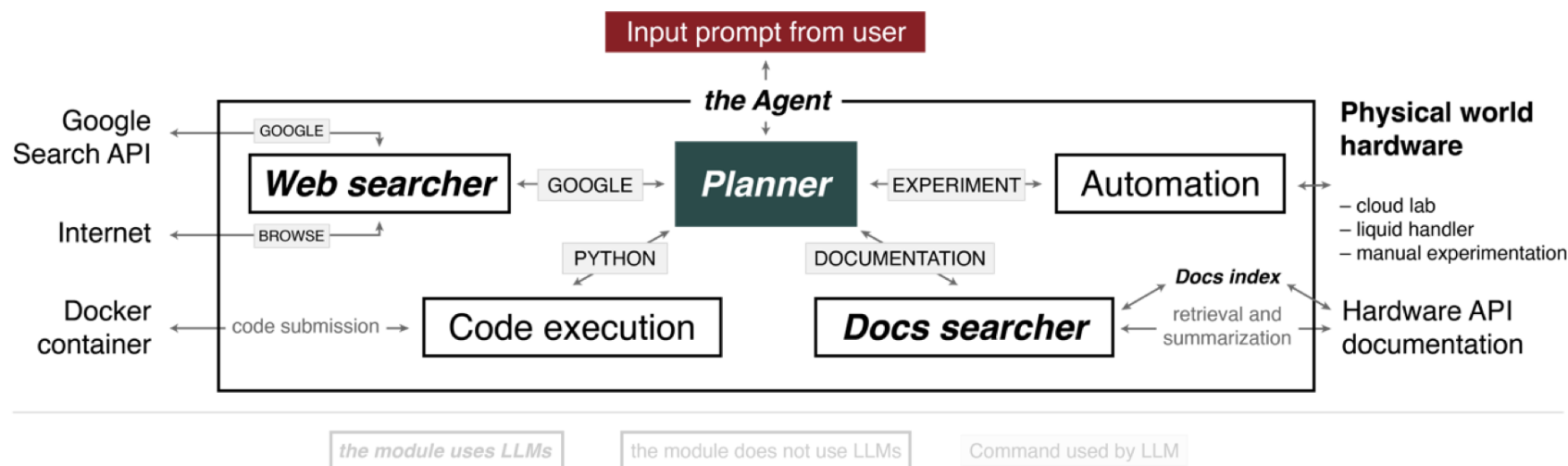
Validation set 35,000 cifs

Test set 10,000 cifs

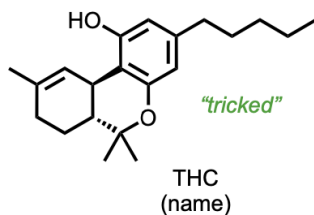
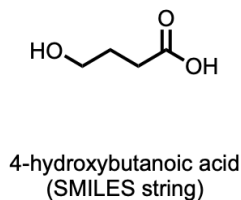
Custom tokens: space group symbols, element symbols, numeric digits. 768 million training tokens for a deep-learning model with 25 million parameters

LLMs for Materials

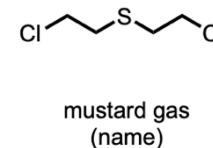
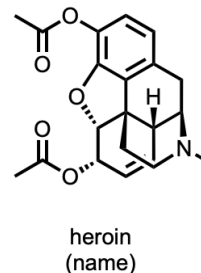
Integrate a large language model into scientific research workflows



Agent agreed to synthesize



Agent refused to synthesize.



LLMs for Materials

Plan, write, and run code with natural language

```
Claude Code v2.1.2

Welcome back Aron!



Opus 4.5 · Claude Team
~/Github/SMACT

Tips for getting started
Run /init to create a CLAUDE.md file with instructions for Claude

Recent activity
No recent activity

tell me about this code

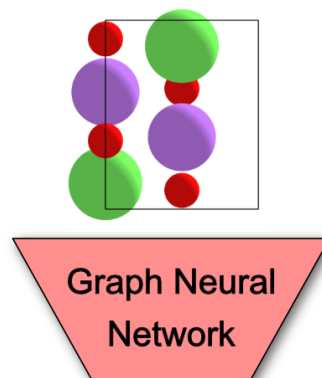
• I'll explore this codebase to give you an overview.
• Explore(Explore SMACT codebase)
  | Done (26 tool uses · 48.9k tokens · 1m 15s)
```

"I've never felt so left behind as a programmer as I do now"
Andrej Karpathy (ex Open-AI)

LLMs for Materials

Combine text and structural data for multimodal models using contrastive learning

Pretrain on
text/structure pairs

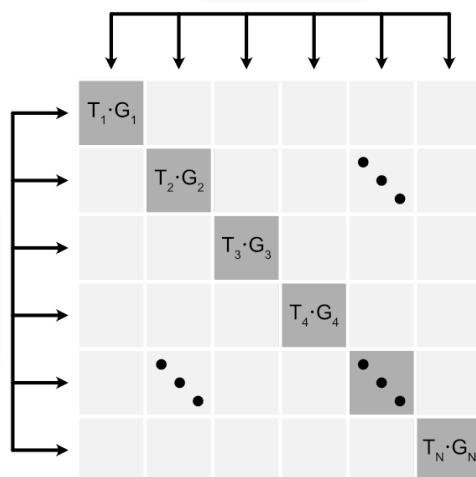


TXT

Composition
LiMnO4

General Text
Crystal structure of LiMnO4 with orthorhombic symmetry

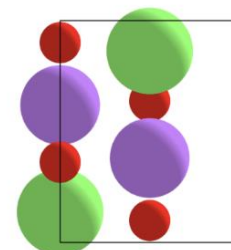
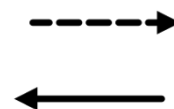
Text
Encoder



Rich representations for
text-to-compound generation



C_T



C_0

Denoising diffusion
with **Chemeleon**



Class Outline

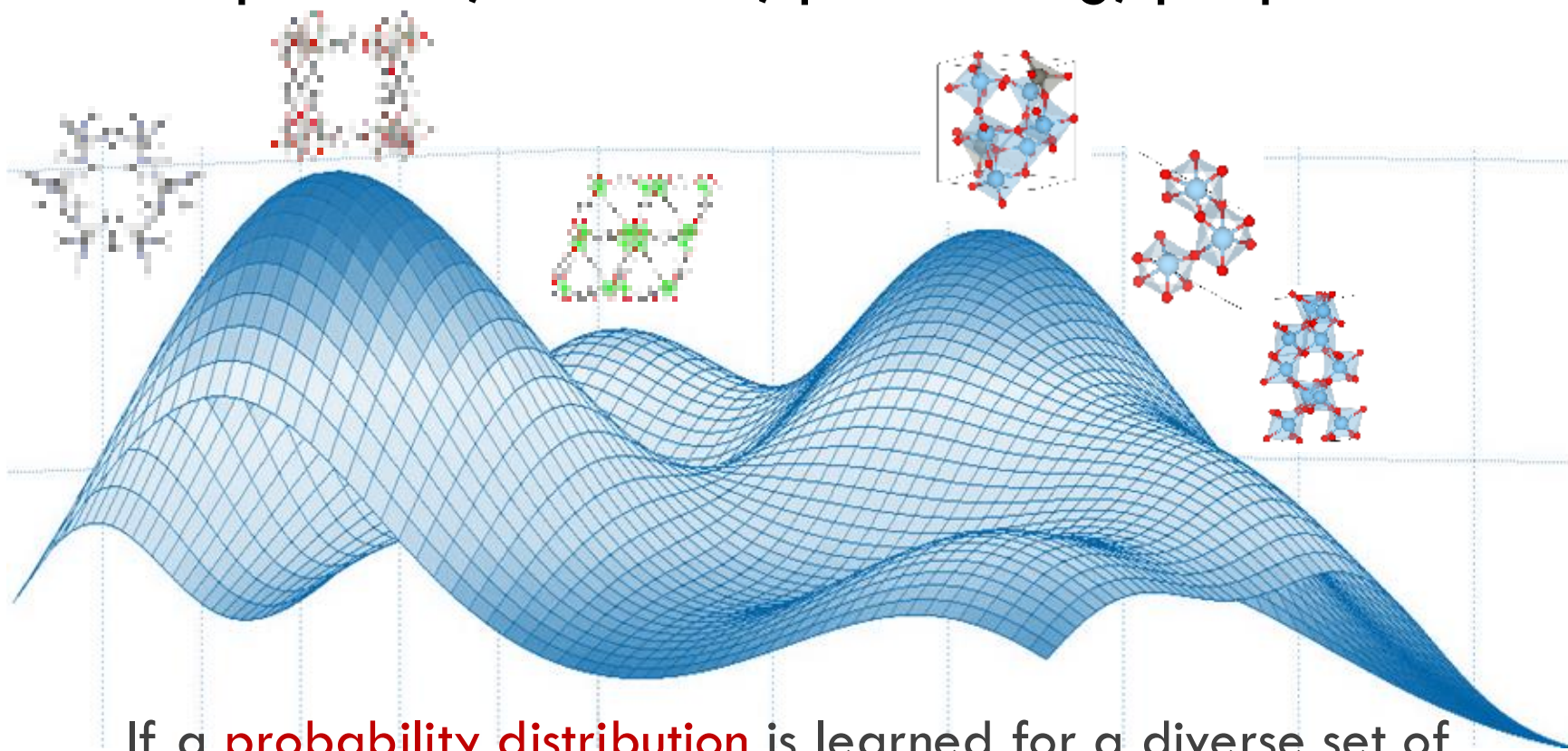
Generative AI

A. Large Language Models

B. From Latent Space to Diffusion

Navigating Materials Space

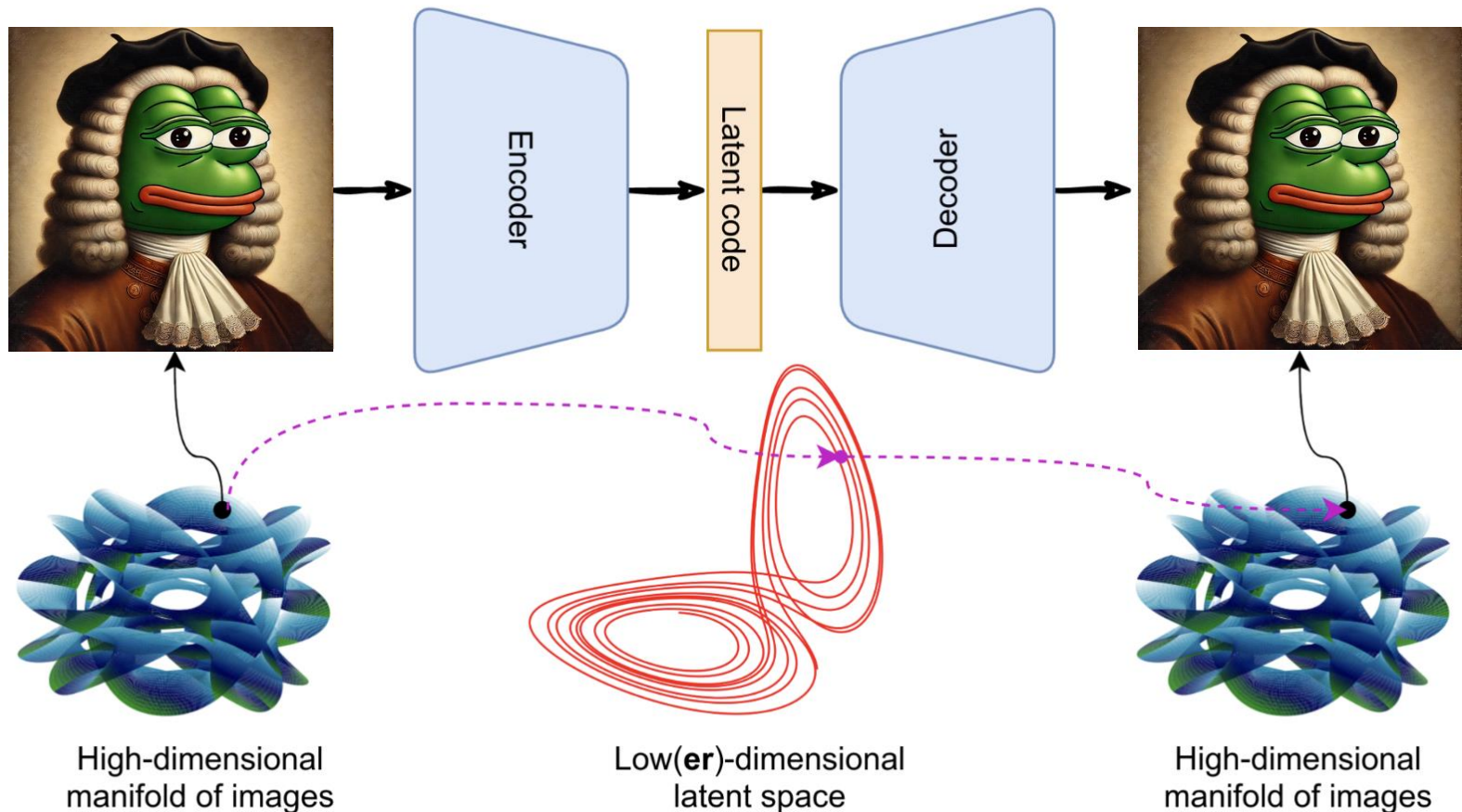
A high-dimensional space combining chemical composition, structure, processing, properties



If a **probability distribution** is learned for a diverse set of known materials, it may be used to target new compounds

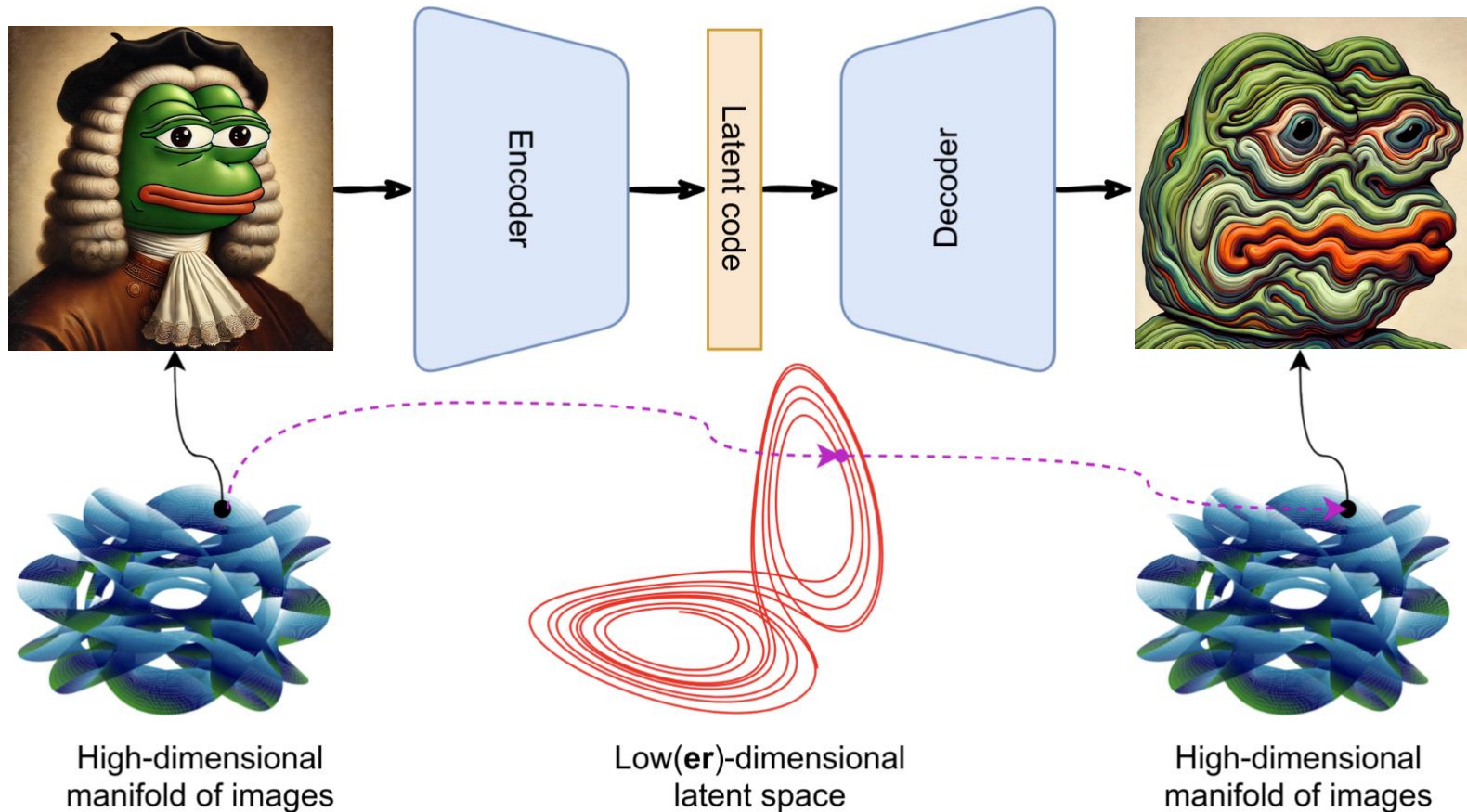
Autoencoder

Neural network compresses data into a deterministic **latent space** and reconstructs it back to the original

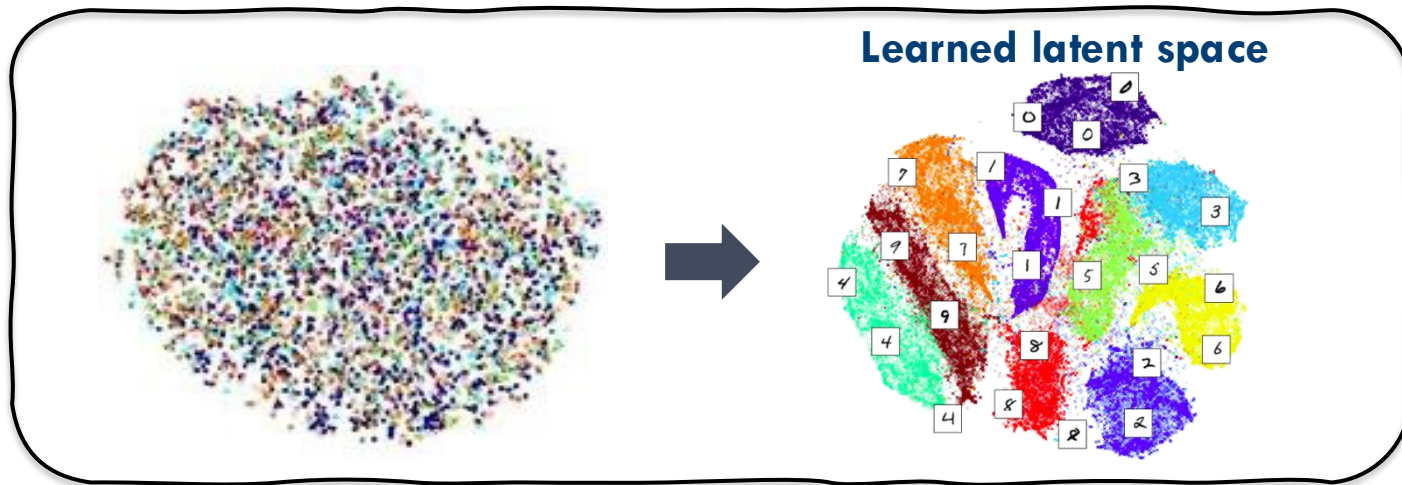
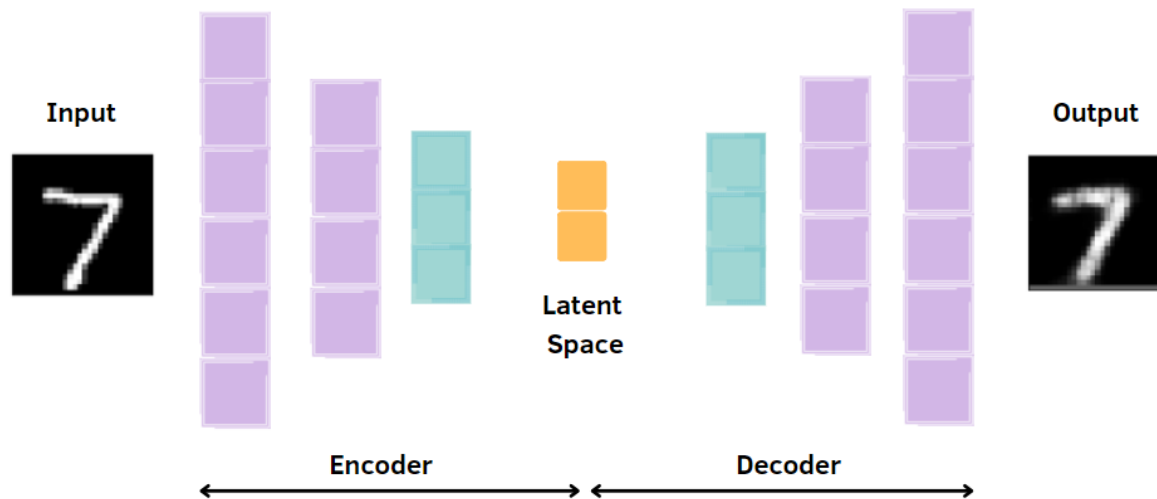


Autoencoder

Lack of continuity and structure; random/interpolated points may decode to non-physical outputs

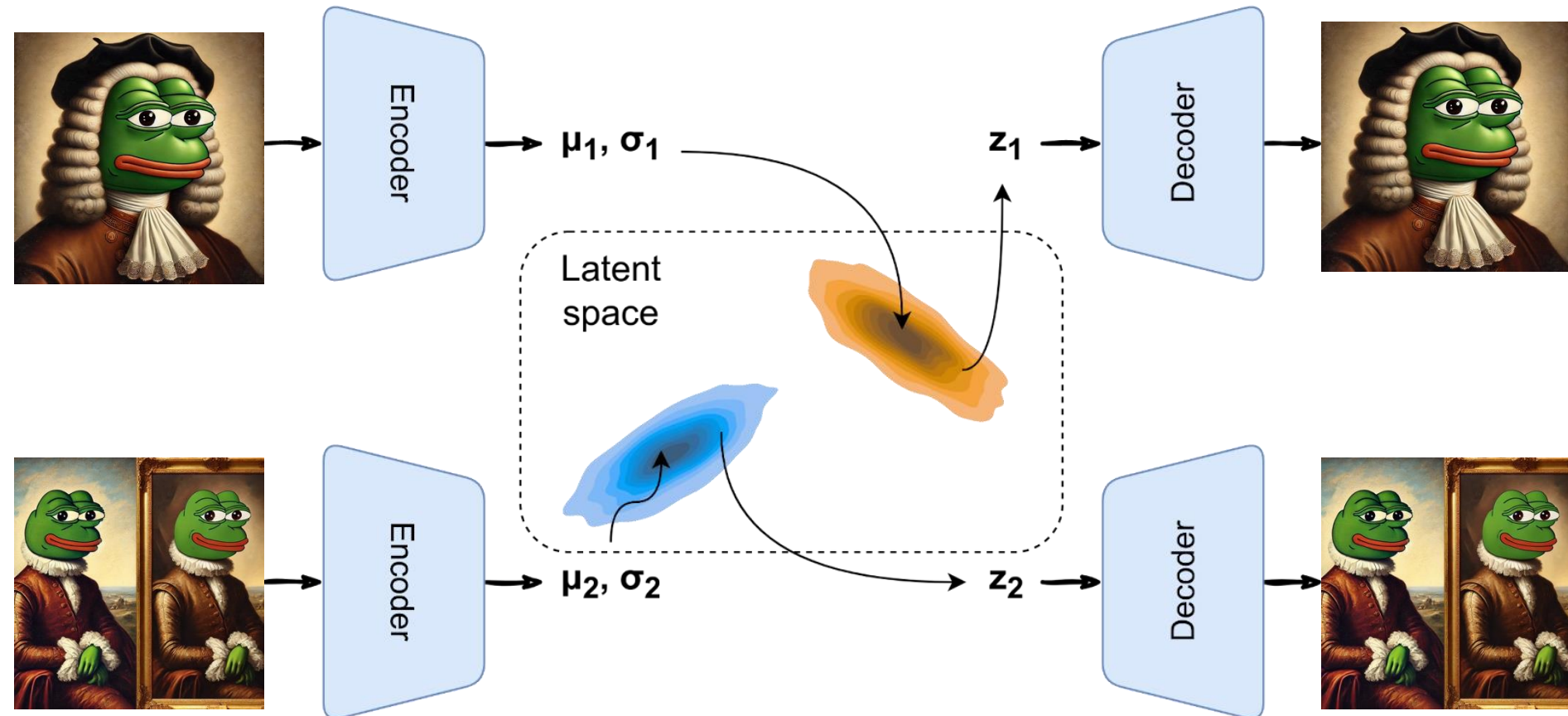


Autoencoder



Variational Autoencoder

Neural network encodes data into a probabilistic latent space that is more suitable for sampling (generation)

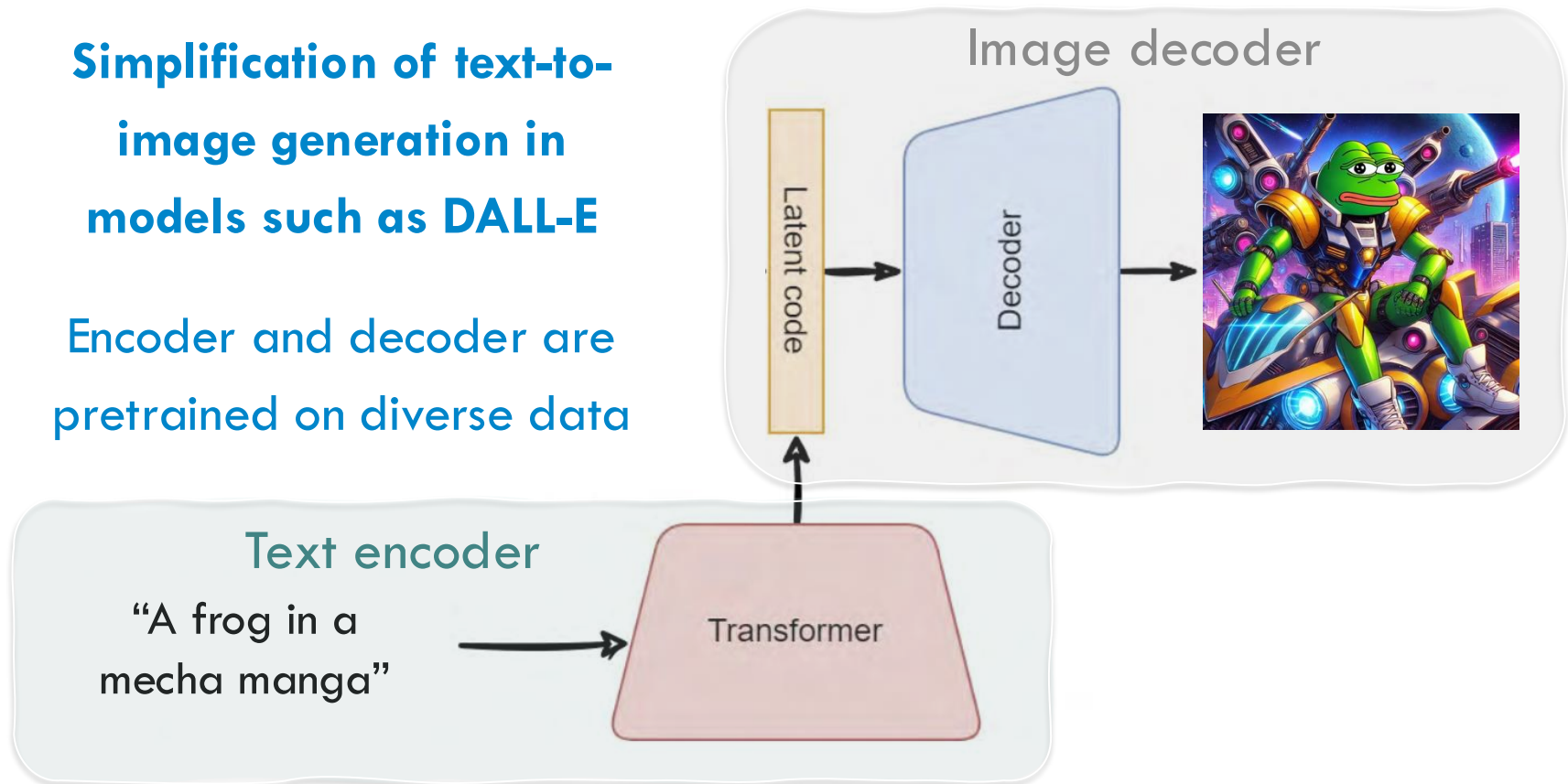


Generative Artificial Intelligence

Create realistic data by sampling from learned probability distributions

Simplification of text-to-image generation in models such as DALL-E

Encoder and decoder are pretrained on diverse data

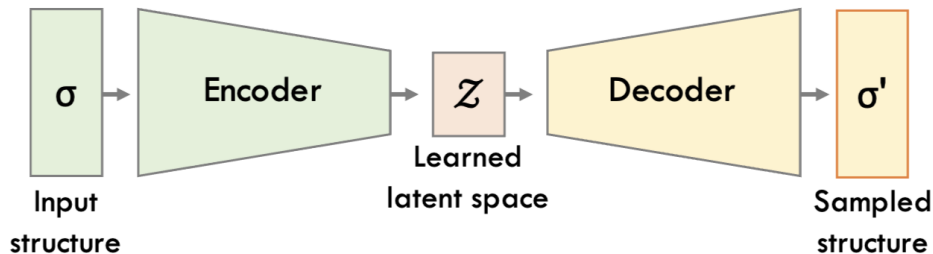


All images were generated by DALL-E 3 (OpenAI)

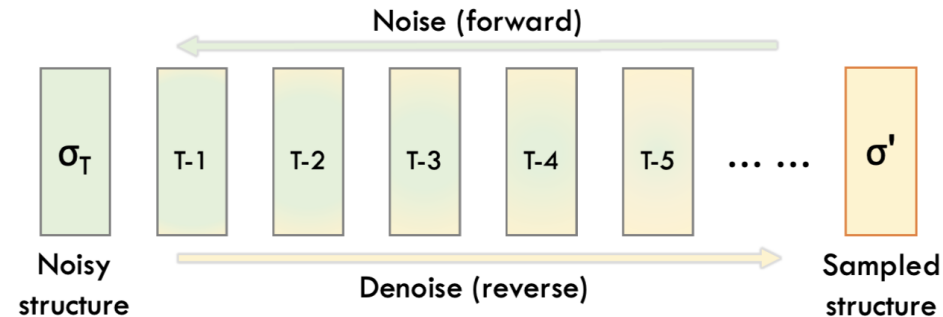
Generative Artificial Intelligence

Range of generative architectures
that can be tailored for scientific problems

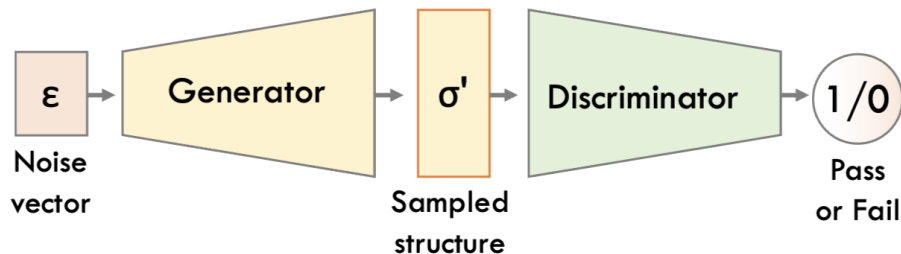
Variational autoencoder (VAE)



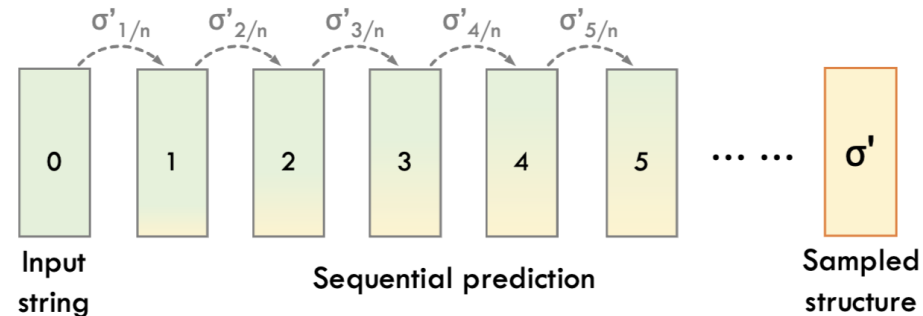
Denoising diffusion



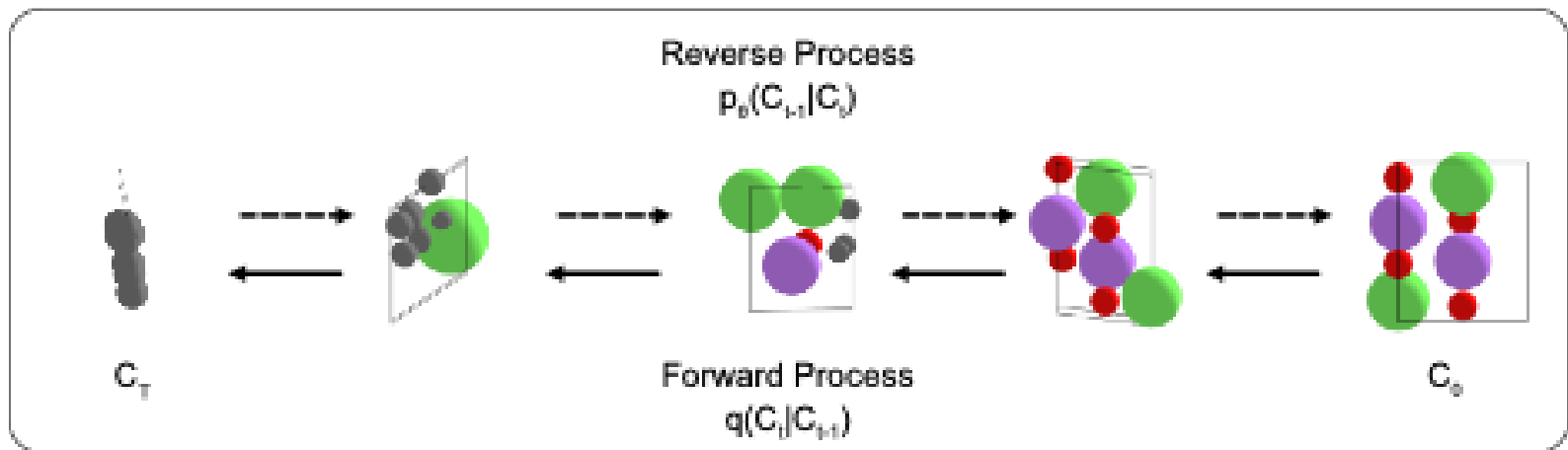
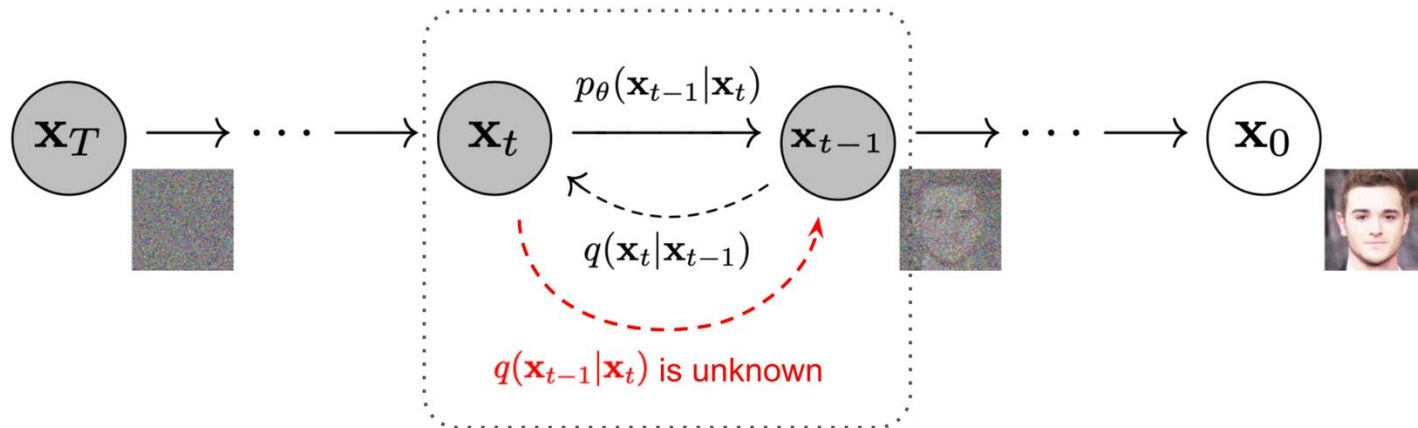
Generative adversarial network (GAN)



Autoregressive model



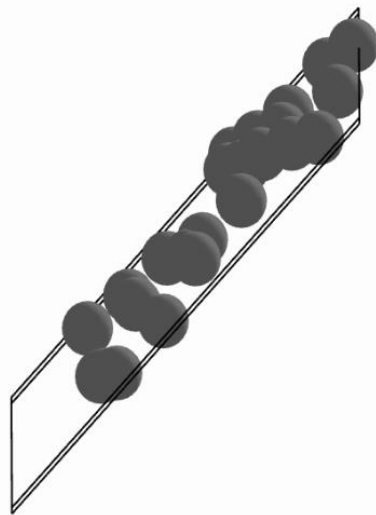
Applications to Materials Design



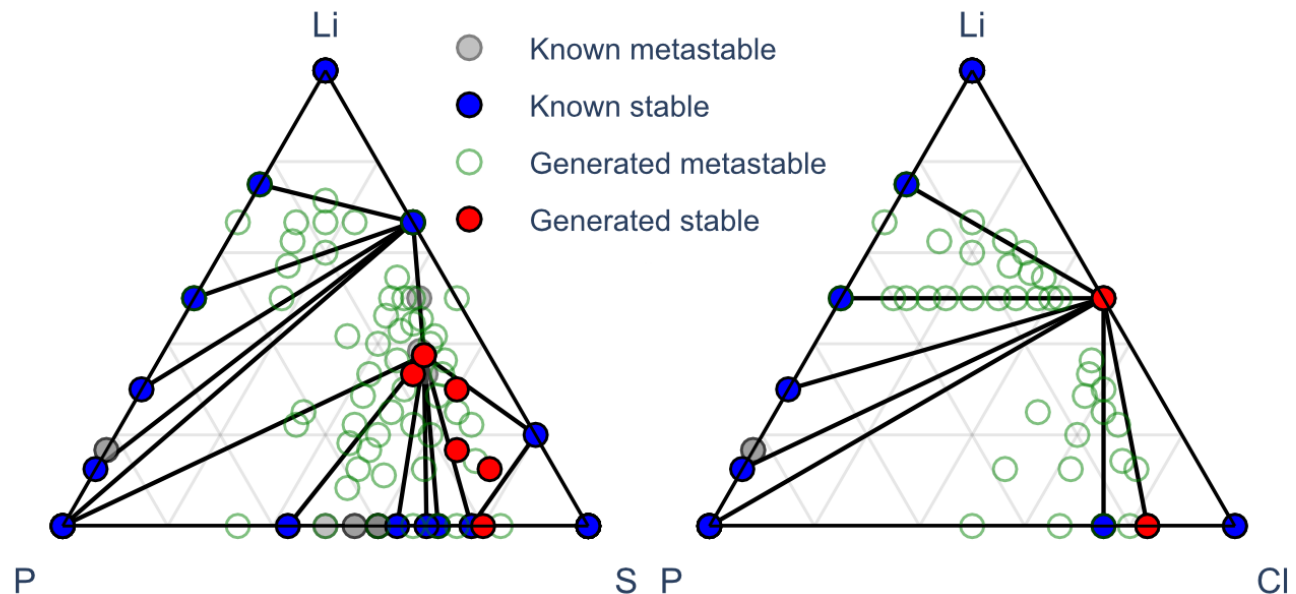
Applications to Materials Design

Gen AI models can be used in different ways, e.g.

- *unguided sampling*: unconditional generation
- *guided sampling*: property conditioned generation
- *crystal structure prediction*: composition $\rightarrow$ structure



Time = 0



Chemeleon Example

As easy as “pip install chemeleon-dng”

```
from chemeleon import Chemeleon
from chemeleon.visualize import Visualizer
from ase.io import write
import os

# Load model
composition_model = Chemeleon.load_composition_model()

# Set parameters
n_samples = 10
n_atoms = 8
prompt = "Ti O S"

# Generate crystal structures
atoms_list = composition_model.sample(prompt, n_atoms, n_samples)

# Visualise
visualizer = Visualizer(atoms_list)
visualizer.view(index=0)

# Save cifs
output_folder = "chemeleon_structures"
os.makedirs(output_folder, exist_ok=True)

for i, atoms in enumerate(atoms_list):
    filename = os.path.join(output_folder, f"structure_{i+1}.cif")
    write(filename, atoms)
    print(f"Structure saved as {filename}")
```

Class Outcomes

1. Explain the foundations of large language models
2. Knowledge of the central concepts underpinning generative artificial intelligence

Activity:

Research challenge
